

Summer Term 2026 – Julián Méndez & Ricardo Langner

Interactive Information Visualization

Exercise: **Assignment 2**

Assignment 2

Tasks and Goals

I. Visualization Tools

Make your own practical experience (w.r.t. workflow, usability, features, ...) using three tools for data visualization

- > *For this task, you will create visualizations using three different tools.*

II. Visualization “Programming”

Gain experience on how to create visualizations using GenAI chatbots

- > *For this task, you will try to develop a visualization with the help of AI.*

III. Interactions and Tasks

Learn how to describe and relate visualization techniques and visualization tasks to each other

- > *For this task, you will analyze and reflect on a specific visualization technique and its characteristics.*

Assignment 2

I. Visualization Tools

1. Choose and download a sample data set from <https://public.tableau.com/app/learn/sample-data>
2. Access and try out the three tools.
 - RAWGraphs, Datawrapper, Tableau Public
 - *Note that you will have to create an account for some tools. You can use your anonymous ZIH email address. Check your ZIH profile at <https://selfservice.tu-dresden.de/profile/> and look out for your alias, such as user.***@tu-dresden.de. In case of reservations, please contact us in time!*
3. With each tool, create 1-2 useful charts for the data you have selected
 - Upload all or a portion of the dataset and create your chart
 - *Note that you are free to either create different charts or replicate them for each tool.*

Assignment 2

I. Visualization Tools

4. Create a **structured** text document and collect your results
 - Add images/screenshots of your visualizations, and for Datawrapper and Tableau Public, also URLs to the published charts/dashboards
 - Briefly describe your experience with the tools. Write about how they compare to each other (e.g., support during creation process, level of abstraction, functionalities, etc.).

Assignment 2

I. Visualization Tools

RAW Graphs (<https://www.rawgraphs.io/>)

RAWGraphs

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A free and open source tool for data visualization

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Main Features

Open source
Inspired by and built on top of open source projects. RAWGraphs is open to the community for contributions.

A wide range of charts
Almost 30 visual models to visualize quantities, hierarchies, time series and find insights in your data.

Your data is safe
Even though RAWGraphs is a web app, the data you insert will be processed only by your web browser.

Export and go anywhere
Save your project, or export it as vector or raster image. Edit it within your favourite softwares.

RAWGraphs

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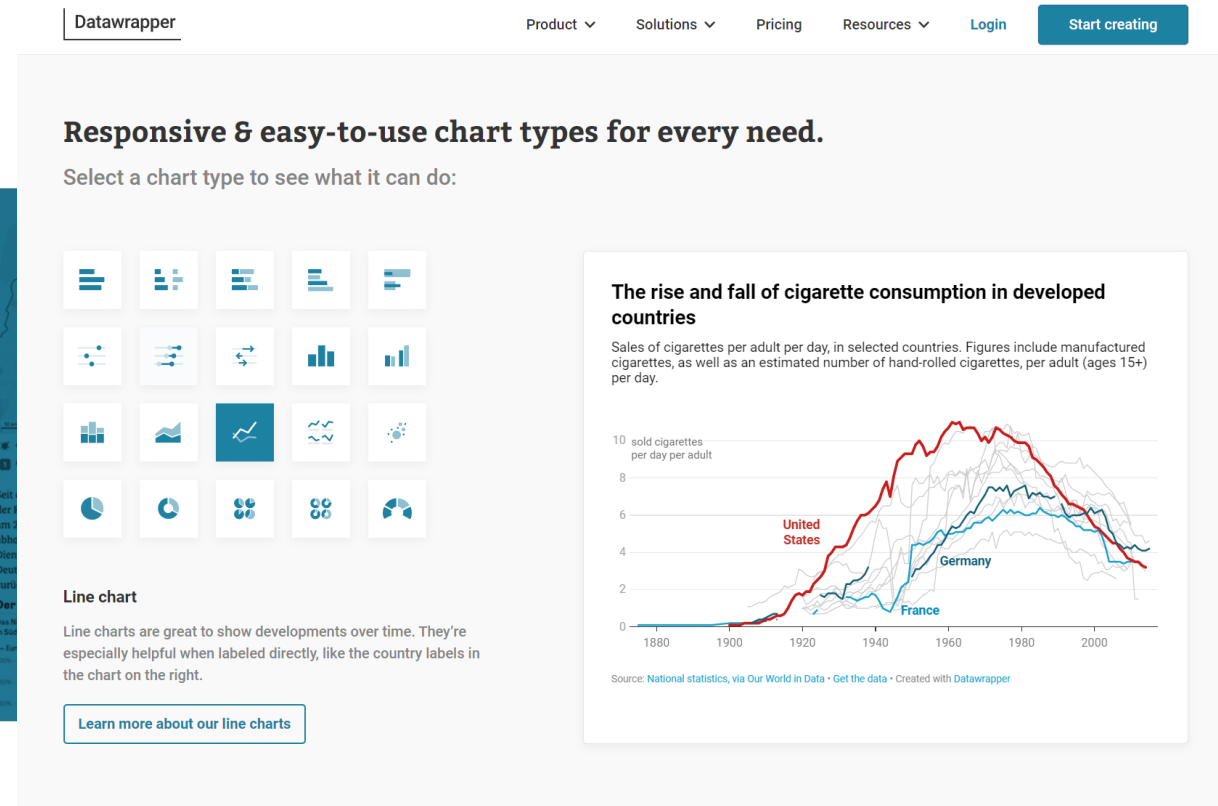
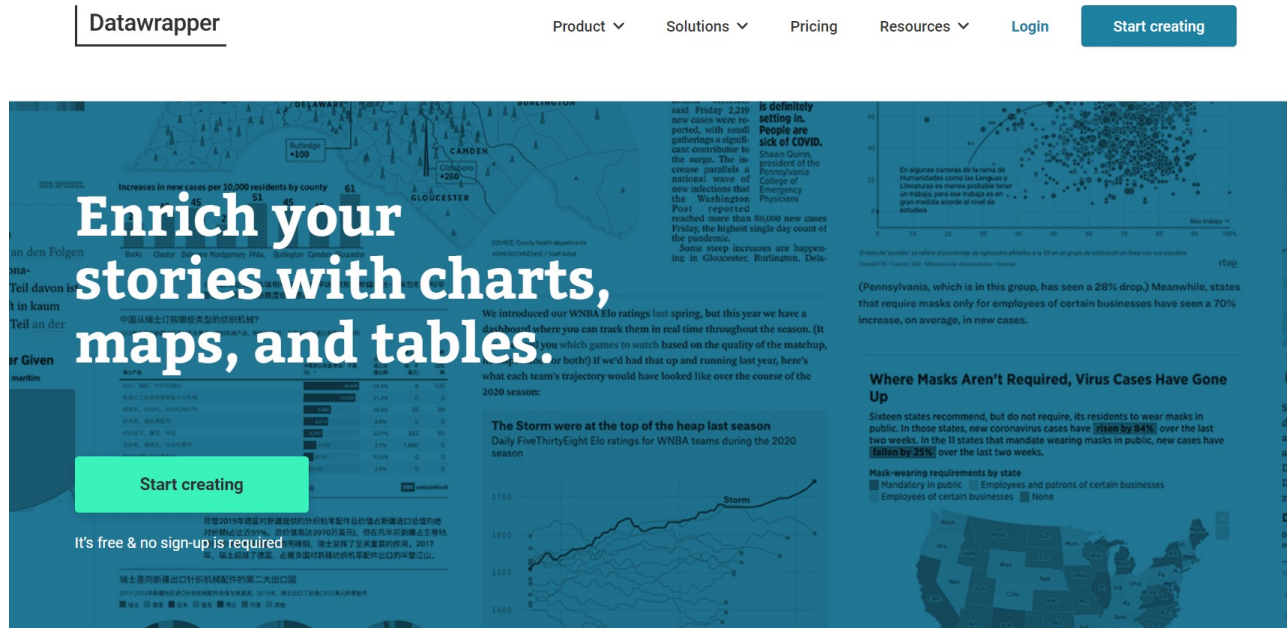
The grid displays a variety of data visualizations: a network graph with nodes and edges, a time series chart with multiple lines, a hierarchical tree, a map visualization, a bar chart, a line chart, a network graph with different styling, a time series chart with area fill, a map visualization, a bar chart, a line chart, and a network graph.

Assignment 2

I. Visualization Tools

Datawrapper (<https://www.datawrapper.de>) – requires an account!

- How-to: <https://academy.datawrapper.de/article/245-how-to-create-your-first-datawrapper-chart>



Assignment 2

I. Visualization Tools

Tableau Public (<https://public.tableau.com>) – requires an account!

- How-to: <https://public.tableau.com/app/resources/learn>

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tableau public Create Resources Sign In

New to Tableau Public? Read our step-by-step guide to getting started on your own data visualization journey.

Welcome to Tableau Public

A free platform to explore, create, and publicly share data visualizations online.

[Sign Up for Tableau Public](#) [Learn More](#)

Viz of the Day

See a new viz featured every weekday

	1 Strongly Disagree	2 Disagree	3 Neutral	4 Agree	5 Strongly Agree
Design	2K	1K	1K	1K	1K
Students	2K	1K	1K	1K	1K

Can You Create a Jittered Bar Chart?

Lorna Brown

Participate in the data viz community-led project Workout Wednesday #WOW23 and learn how to create a jittered bar chart with this #VizOfTheDay created by Tableau Visionary Lorna Brown.

☆ 31 👁 5,097

salesforce

tableau public Create Resources Sign In

New to Tableau Public? Read our step-by-step guide to getting started on your own data visualization journey.

Resources

Explore how-to videos, sample data, and community resources to help you get started or to take your skills to the next level.

[Learn](#) [Sample Data](#) [Community Resources](#)

Executive Overview - Profitability (All)

Monthly Sales by Segment - States: All

Monthly Sales by Category - States: All

Currently Playing: Get Started (0:22)

Assignment 2

II. Visualization “Programming”

1. Choose and download a sample data set from <https://public.tableau.com/app/learn/sample-data>
2. Select any GenAI chatbot.
 - *Note that you are free to use any AI tool as support, be it a chatbot in the browser or in the code editor. For some you will have to create an account. You can use your anonymous ZIH email address again (see task 1). In case of reservations, please contact us in time!*
3. Develop a basic visualization (preferably bar chart, line chart, or scatterplot) for your data using the visualization programming library D3 (<https://d3js.org>). Add screenshots of your result.
4. Briefly describe your experience with this task. Was it easy, complicated, successful, ...?
5. Briefly compare and reflect on your experience with the approaches from Task I and Task II.
What-you-configure-is-what-you-get versus What-you-describe-is-what-you-hope-to-get

Assignment 2

III. Interactions and Tasks

1. Remember what you have learned about a parallel coordinates plot (also called PCP, ||-plot) in the lecture. You can also watch <https://youtu.be/fU5YYdfjUNw> for an introduction.
2. Read Yi et al.'s 2007 article *"Toward a Deeper Understanding of the Role of Interaction in Information Visualization"*, focus on its section 4*.
 - PDF in OPAL or <https://faculty.cc.gatech.edu/~stasko/4460/Reading/yi-tvcg07.pdf>

* Note, we also recommend section 3 of Brehmer & Munzner's 2013 article *"A Multi-Level Typology of Abstract Visualization Tasks"* and their website as additional in-depth reading on the topic of InfoVis tasks.

- PDF in OPAL or https://www.cs.ubc.ca/labs/imager/tr/2013/MultiLevelTaskTypology/brehmer_infovis13.pdf
- Brehmer & Munzner's Website: <https://www.cs.ubc.ca/labs/imager/tr/2013/MultiLevelTaskTypology/>

Assignment 2

III. Interactions and Tasks

3. Reflect on how a PCP supports Yi's categories of interaction. For each of the seven categories (select, explore, reconfigure, encode, abstract/elaborate, filter, connect), briefly describe if the category is supported and how it is supported.
4. If you find that one of the categories is not or not sufficiently supported by PCP, briefly sketch and describe a possible extension (functionality) of PCP to overcome the supposed shortcoming.

user's intent in performing the interaction. We describe each category to provide a definition of what it means and we also include exemplary individual interaction techniques that fall within that category.

- *Select*: mark something as interesting
- *Explore*: show me something else
- *Reconfigure*: show me a different arrangement
- *Encode*: show me a different representation
- *Abstract/Elaborate*: show me more or less detail
- *Filter*: show me something conditionally
- *Connect*: show me related items

4.1 Select: mark something as interesting

Select interaction techniques provide users with the ability to mark a data item(s) of interest to keep track of it. When too many data items

Image source: from the article <https://doi.org/10.1109/TVCG.2007.70515>

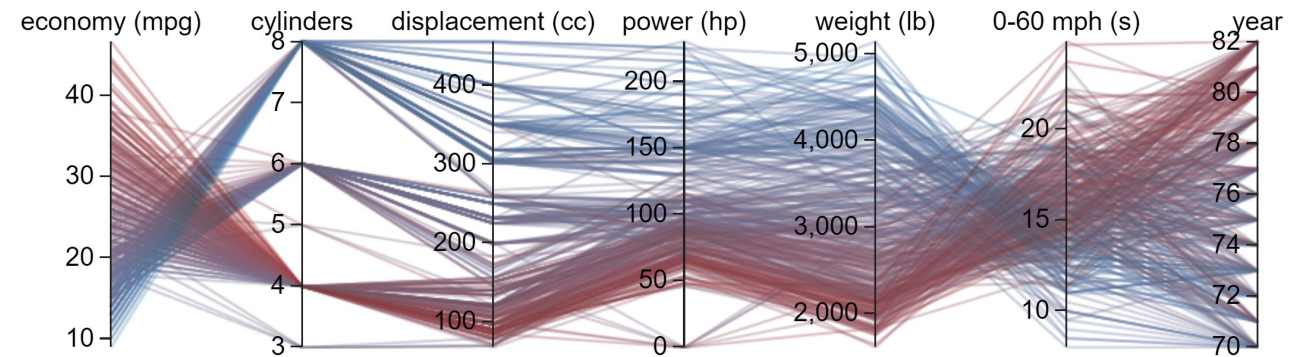


Image source: <https://syntagmatic.github.io/parallel-coordinates/>

Assignment 1

Summary of your tasks

In your team ...

1. Go through these slides again and discuss the tasks, maybe distribute work.
2. Create some structured text document.
3. Work on the tasks I, II, and III and collect your results in your text document.
4. Discuss your results in the team.
5. Upload your results to the submission folder (remember the file naming scheme from the introductory slides!).
6. Bring your results with you (the document, notes, screenshots, code, ...) to the on-site exercise on May 22, because we will discuss/reflect about them in group work.

Exercise Interactive Information Visualization

Thank you!

Any questions?

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